Important Question ADA for 2025-2026 4th SEM CSE/AD/DS/CI/AI

1. Define Algorithm explain steps involved in algorithm design and analysis process with a neat diagram.

2. Explain order of growth with an example order the following to their order growth from the lowest to highest!N-2, logN,N3,2 power N, N cube, N square, NlogN

3. Write an Algorithm to linear search find out time complexity for an algorithm.

4. Define algorithm. Explain asymptotic notations Big Oh, Big Omega and Big Theta. Notations.

5. Explain the general plan for analyzing the efficiency of a non-recursive algorithm. Suggest a non-recursive algorithm to find maximum element in the list of n numbers. Derive its efficiency .

6. f t1(n) O(g1(n)) and t2(n) O(g2(n)), then show that t1(n) + t2(n) O(max{g1(n), g2(n)})

7. ith neat diagram explain different steps in designing and analyzing an algorithm .

Explain the general plan for analyzing the efficiency of recursive algorithm. Suggest a recursive algorithm to find factorial of number. Derive its efficiency.

8. With the algorithm derive the worst case efficiency for Bubble sort.

9. Explain General plane for the non recursive algorithm suggest an algorithm to find which the element in an array are unique ,analyse its efficiency.

I)Solve the following Recurrence relation
X(n)=x(n-1) + 5 for n>1 , x(1)=0

ii)4. Solve the following Recurrence relation
10.X(n)=3x(n-1) for n>1 , x(1)=4

11.Using Bubble sort letter word QUESTION sort in alphabetical order.

Explain the concept of divide and conquer. Design an algorithm for merge sort and derive its time complexity.

12.Design an insertion sort algorithm and obtain its time complexity. Apply insertion sort on these elements. 25,75,40,10,20,

13.Explain Strassen's matrix multiplication and derive its time complexity

14).Solve the following Recurrence relation
T(n)=0 if n=0T(n)=1+t(n-1) otherwise.

ii)Solve the following Recurrence relation
14.X(n)=3x(n-1) for n>1 , x(1)=4

15.Explain and Write Merge sort algorithm. Find out Time Complexity of Merge sort using . i)Master Theorem ii)Substitution Method.

16.Design an insertion sort algorithm and obtain its time complexity. Apply insertion sort on these elements. 25,75,40,10,20,

17.Apply strasssen's algorithm to multiply the given two matrices .

A= 1 2 B= 8 7

5 6 and 1 2

18.Design an algorithm for quick sort algorithm. Apply quick sort on these elements. Sort array list using using Quick Sort EXAMPLE .

19. Find out time complexity. Determine the number of comparisons researching for the pattern CSE using the text VTUCSECOURCE.
20.Sort array list using using Quick Sort
20. 65,70,75,80,85,60,55,50,45.
Solve the knapsack problem obtain the optimal solution using Broute force technique. Given problem
21. M=10, n=4 ,w1,w2,w3,w3=7,3,4,5 and Profit P1,p2,p3,p4=42,12,40,25
22. Define Exhaustive Search approached with Brute Force Technique to solve the Travelling Salesman Problem with an example. Write efficiency class of this approach.
Consider the Algorithm

```
SECRET(n, a):
Minval<-0
        for i=0 to n-1 do
            If(a[i]<minval )
              Minval=a[i]
           If(a[i]>maxval )
              Maxval=a[i]
```

i) What does this algorithm compute.

ii) What is the basic operation.

iii) How many times the basic operation executed.

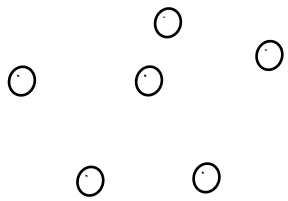
iv) What is the efficiency class of this algorithm.

v) write time complexity with notation

23.MYSTRY(n):
S<--0
for i=0 to n do
s=s+i*I
return s

vi) What does this algorithm compute.
vii) What is the basic operation.
viii) How many times the basic operation executed.
Iv)What is the efficiency class of this algorithm.

a) 24.Find the Topological order of the given graph by using source removal method.



25 Using Selection sort letter word EXAMPLE sort in alphabetical order.

